

Cover Letter

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Editor npj Microgravity Nature Portfolio

Dear Editor,

We are pleased to submit our manuscript entitled “Light quality modulates spaceflight stress decoding with cell-type asymmetry in tomato” for consideration in npj Microgravity.

This study addresses a fundamental question in spaceflight plant biology: how do growth conditions modulate the plant’s transcriptional response to the spaceflight environment, and which established stress response programs does spaceflight most resemble? Using RNA-seq data from the OSD-767 mission, in which tomato plants were grown aboard the International Space Station under red or blue LED illumination, we employed a 2×2 factorial design to decompose transcriptomic variation into light effects, spaceflight effects, and their interaction. We further applied PhysioSpace stress-pattern decoding to map the spaceflight transcriptomic response onto a compendium of 20 Arabidopsis stress-response signatures. Our findings advance the field in four key respects:

1. **Compensatory dynamics between light quality and spaceflight.** We show that the Light $\times$ Condition interaction comprises 25% of the leaf transcriptome (5,042 genes), with

a strong negative correlation ($r = -0.84$) between light and interaction log-fold-changes. This compensatory pattern—where pathways pre-activated by one light condition show attenuated spaceflight responses—has not been previously described and has important implications for interpreting existing spaceflight transcriptomic studies that did not control for light quality.

2. **Cell-type-specific radial asymmetry in root spaceflight responses.** Using TRAP-seq markers to deconvolve bulk transcriptomic data, we reveal that root cell types show a radial gradient of spaceflight sensitivity, with outer cortex (Cortex_ACT2) showing extreme enrichment for interaction-up genes ($OR = 26.9$, $p = 1.84 \times 10^{-12}$) and inner cell types showing balanced responses. This spatial patterning of spaceflight sensitivity has not been previously reported.
3. **Stress-pattern decoding reveals tissue-specific defense priming.** Using PhysioSpace to map spaceflight transcriptomes onto a compendium of 20 Arabidopsis stress-response signatures, we show that spaceflight suppresses light-related stress programs in leaves (Light flight effect: -27.2 to -32.2) while activating biotic/hormone defense programs in roots (Biotic.Hormone flight effect: $+14.2$ under red LED, $+58.4$ under blue LED). The root biotic/hormone response shows the strongest light-quality interaction of any stress axis (interaction = -44.2), driven by dramatic blue-LED amplification of defense signaling. This stress-pattern-level perspective connects the gene-level compensatory dynamics to established stress biology and provides a new framework for interpreting spaceflight transcriptomic data.

4. Practical implications for ISS growth chamber design.

Our finding that blue light partially rescues spaceflight suppression of photosynthesis genes while hyperactivating root defense signaling, and that red light amplifies hormone stress signaling, directly informs the design of light regimes for spaceflight plant growth systems. We propose that dynamic light spectra balancing these competing effects may optimize plant productivity in microgravity.

This work is well-suited for npj Microgravity because it combines rigorous statistical methodology (factorial interaction models with LFC shrinkage), cell-type-resolved analysis, and a novel stress-pattern decoding approach (PhysioSpace) that provides pattern-level interpretation directly connecting novel spaceflight responses to established stress biology. The study integrates four complementary analyses—asymmetric regulation, cell-type-resolved pathway projection, light-condition interaction, and PhysioSpace stress-pattern decoding—into a single coherent narrative with actionable recommendations for spaceflight hardware design.

All data are available from NASA GeneLab (OSD-767), and analysis code will be deposited in a public repository upon acceptance. This manuscript has not been published previously and is not under consideration elsewhere. No preprint has been posted.

We believe this work will be of broad interest to the npj Microgravity readership, spanning spaceflight biologists, plant physiologists, and bioinformaticians. We look forward to your consideration.

Sincerely,

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